Requirements-based testing of neural networks.

A critical review of RBT4DNN, its empirical evidence, and two focused additional analyses.

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Three questions guide the argument.

01 · NEED

Why do neural-network tests need requirements?

PROBLEM + BACKGROUND

02 · METHOD

How does RBT4DNN generate and evaluate targeted tests?

APPROACH + METHODOLOGY + RESULTS

03 · EVIDENCE

How convincing—and scalable—is the resulting evidence?

CRITIQUE + ADDITIONAL ANALYSES

These questions follow the seminar brief while keeping one continuous evidence argument.

High average accuracy does not answer a specific requirement.

CONVENTIONAL TESTING

Known behavior

A specification maps an input condition to an expected output. Test oracles can check the result directly.

DNN TESTING

Learned behavior

The model is inferred from examples. Global accuracy summarizes a distribution, not every meaningful scenario.

THE MISSING LINK

Semantic scenarios

Developers need evidence for statements such as "a very thick 7 must still be classified as 7."

GLOBAL METRIC
How often is the complete test set correct?

→

SEMANTIC SLICE
Which rare inputs does the score contain?

→

REQUIREMENT VERDICT
Does the model satisfy this specific contract?

[1] §1–§2.5 · "model under test" means the evaluated neural network

Existing generators rarely start from the requirement itself.

Neuron coverage, image transformations, and search-based generators can produce many tests—without proving that those tests represent the scenario a stakeholder cares about.

INPUT	Requirement	Structured natural language defines a semantic precondition and an expected outcome.
METHOD	RBT4DNN	Fine-tune a generator on images that satisfy that precondition.
OUTPUT	Targeted tests	Run the learned component and evaluate the requirement-specific oracle.

[1] §1, §2.5 and §3 · authors claim the first requirement-driven DNN test-input generator

Four domains test whether one workflow survives increasingly ambiguous semantics.

MNIST · 7 REQUIREMENTS

MNIST artifact samples
Source-linked in interactive deck

Digits

Measured height, thickness, and lean. A controlled starting point.

Output: digit class

CELEBA-HQ · 7 REQUIREMENTS

CelebA-HQ artifact samples
Source-linked in interactive deck

Faces

Ambiguous attributes labeled by visual question answering or humans.

Output: eyeglasses decision

SYNTHETIC DRIVING (SGSM) · 7

SGSM artifact samples
Source-linked in interactive deck

Driving

Scene relations linked to steering and acceleration constraints.

Output: numeric control

IMAGENET · 4 REQUIREMENTS

ImageNet artifact samples
Source-linked in interactive deck

Animals

Morphology linked to allowed classes in the WordNet taxonomy.

Output: class set

[1] Table 2 and §4.1.1–§4.1.3 · samples linked from the pinned public artifact

MNIST M3 artifact samples
Source-linked in interactive deck

A requirement defines both relevance and correctness.

If the digit is a 7 and is very thick, then the model shall label it as 7.

PRECONDITION	POSTCONDITION
Which images count as relevant tests?	Which model outputs count as acceptable?

[1] Table 2, M3 · a finite passing suite cannot prove the universal requirement

RBT4DNN turns one requirement into a targeted test campaign.

01 · LABEL

→

Pixels → terms

Attach domain facts such as very thick, eyeglasses, or same lane.

02 · FILTER

→

Terms → subset

Keep original images satisfying one requirement precondition.

03 · ADAPT

→

Subset → LoRA

Fine-tune FLUX with a small adapter for that precondition.

04 · TEST

Images → evidence

Generate tests, run the model under test, and check the required output.

M3 LABELS	FILTERED SUBSET	TARGETED GENERATOR	EVIDENCE
digit = 7 · thickness = very thick	only matching MNIST images	one M3 LoRA adapter	precondition match + output pass or fail

[1] Fig. 2 and §3 · symbols are translated into an explicit input/output contract

Glossary terms make visual requirements operational.

GLOSSARY TERM

A **named visual property** that can be assigned to an image: “very thick,” “same lane,” or “eyeglasses.”

01 · DEFINE

Authors specify the glossary

Requirement concepts are manually turned into a finite vocabulary of visual properties.

TERM CREATION IS NOT AUTOMATIC

02 · LABEL ORIGINAL DATA

Labels come from domain-specific signals

MNIST measurements · SGSM scene graphs · MiniCPM visual Q&A for CelebA-HQ and ImageNet.

CELEBA ALSO HAS HUMAN LABELS

03 · USE THE LABELS

Filter first; check later

Labels select the LoRA training subset. Separate binary classifiers then check glossary terms on generated images.

PRECONDITION MATCH · RQ1 / RQ3

Boundary: separate training labels and evaluation classifiers reduce direct reuse, but neither constitutes independent semantic ground truth.

[1] §3.2 and §4.1.6 · mean glossary-classifier accuracy 94.5%; C4 is 70.8%

One filtered precondition subset trains one targeted LoRA.

FROZEN BACKBONE

$$W$$

FLUX.1-dev: a pretrained 12-billion-parameter image generator.

+

TRAINABLE UPDATE

$$BA$$

Small low-rank matrices adapt the model to one filtered subset.

BENEFIT	Generate many new candidates inside a rare requirement region
EVIDENCE SOUGHT	Relevance, distributional plausibility, and semantic variation
NOT ESTABLISHED	Non-memorization or systematic coverage within that region

[1] §3.3 and §4.1.4 · LoRA equation simplified; latent-space coverage remains future work

Three comparisons isolate three different effects.

FILTERING · RQ1	Targeted LoRA ↔ all-data LoRA	Does selecting precondition-matching images matter?
ADAPTATION · RQ2–RQ3	Targeted LoRA ↔ prompt-only FLUX	Does fine-tuning improve distributional quality?
TARGETING · RQ1 + RQ4	RBT4DNN ↔ DeepHyperion + transformations	Does requirement guidance improve relevance?

Important: prompt-only FLUX is not an RQ1 comparator; the all-data LoRA and untargeted baselines are dropped after their poor RQ1 precondition match.

[1] §4.1.4–§4.1.6 · no truly comparable requirement-targeted baseline existed

The six RQs validate generated inputs before interpreting model outcomes.

Validate the generated inputs

RQ1–RQ3

RQ1	Consistency	Does the image satisfy the precondition?
RQ2	Plausibility	Is its distribution close to matching training images?
RQ3	Variation	Are other glossary features distributed similarly?

Evaluate the resulting tests

RQ4–RQ6

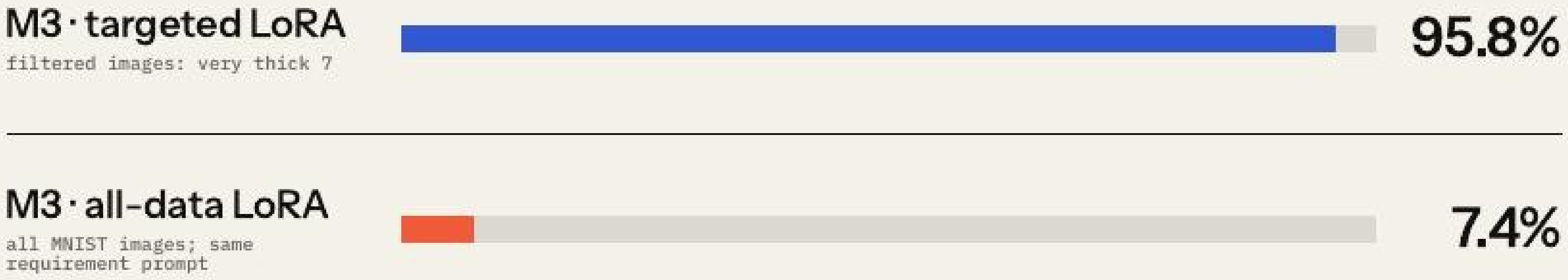
RQ4	Comparison	Are targeted outcomes more informative than baselines?
RQ5	Violations	Do tests reveal credible postcondition failures?
RQ6	Diagnosis	Do failures expose repeatable decision patterns?

[1] §4.1 · input validity is established before model outcomes are interpreted

Targeted LoRAs satisfy the requested precondition 92.1% of the time on average.

92.1%

mean across 21 requirements in MNIST, CelebA-HQ, and SGSM.



METHOD

Every glossary term in the precondition must pass its binary classifier · 100 images × 10 repetitions.

BOUNDARY

The classifiers average 94.5% accuracy; they are measurement proxies, not semantic ground truth.

[1] §4.2.1 and Fig. 3 · prompt-only FLUX is not part of RQ1

Targeted LoRA is much closer for MNIST and SGSM; CelebA-HQ gains are smaller and mixed.

77.09%

reported average KID reduction across Table 3's MNIST M1–M7 and SGSM S1–S7 rows.

COMPARE	**targeted LoRA ↔ prompt-only FLUX**
SAME INPUT	**identical precondition prompt**
CELEBA-HQ	**12.9% automated · 27% human labels**

METHOD

KID compares each generated set with the original training subset matching that precondition · lower means closer.

BOUNDARY

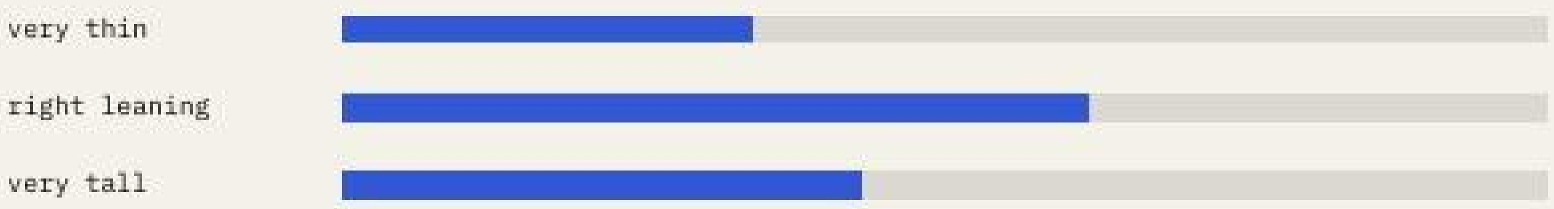
KID is not an image-level realism probability; several reference subsets contain fewer than 300 images.

[1] §4.2.2 and Table 3 · paper prose says M1–M6, but 77.09% matches M1–M7 arithmetic

Generated tests broadly preserve the measured glossary-feature mix.

FILTERED TRAINING SUBSET

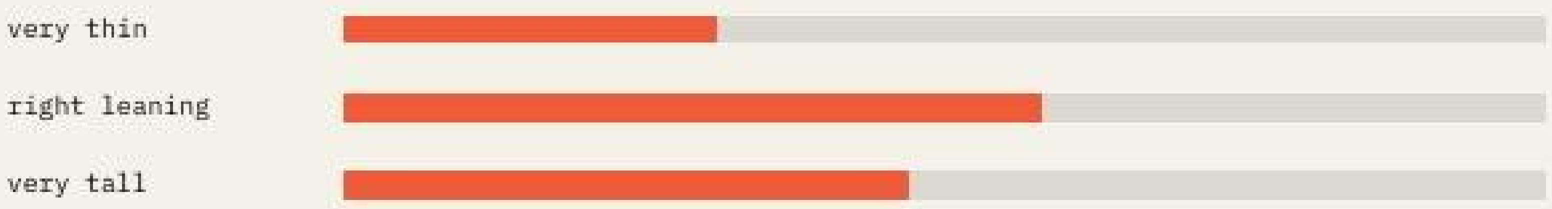
Other glossary features



↔

GENERATED TESTS

Glossary-classifier frequencies



RESULT + METHOD

JS divergence over other glossary-term frequencies: more than 3× closer for MNIST/SGSM; about 32% better for CelebA-HQ.

BOUNDARY

This does not measure pixel-level diversity, memorization, or systematic coverage inside the requirement region.

High pass rates are uninformative when generated images miss the requirement.

DEEPHYPERION

Feature-space search

Recognizable digits, but the search is not conditioned on "very thick."

IMAGE TRANSFORMATIONS

Rotate · blur · translate

Changes existing images without explicitly targeting the named semantic region.

RBT4DNN

Requirement-targeted

Generation reaches M2's "digit 3 and very thick" precondition.

WHAT THE VISUAL SHOWS

DeepHyperion and transformations create recognizable 3s, but not thick 3s; RBT4DNN reaches the named scenario.

INTERPRETATION RULE

First establish input relevance; only then does a classifier pass or failure address M2.

[1] §4.3.1, Fig. 3, Fig. 7 and Table 5 · DeepHyperion and transformations are untargeted

RBT4DNN exposed failures broadly, and the audited subset was mostly valid.

OBSERVED FAILURES

19/21

requirements

At least one postcondition violation was found.

→

AUDIT RULE

11 × 30

sampled failures

Requirements below 90% pass rate received multi-author review.

→

AUDITED QUALITY

3.3%

mean false positives

330 images total; 11.5% worst case.

METHOD

Run the learned component, apply the postcondition oracle, then jointly assess realism and precondition match.

BOUNDARY

The audit supports the reviewed samples—not every generated failure or an internal model-fault localization.

[1] §4.3.2 and Fig. 7 · assessor disagreement on precondition consistency was 1.7%

Failures reveal repeatable patterns—but not their causes.

01

Generate and run

Four zoological requirements against three high-accuracy ImageNet classifiers.

→

02

Collect 1,486 failures

Postconditions ask for a taxonomically compatible prediction, not one exact class.

→

03

Find prediction clusters

Wrong outputs concentrate in a few repeated labels and differ across models.

→

04

Inspect associated images

Background objects, partial animals, and multi-object scenes motivate hypotheses.

EVIDENCE

Prediction frequencies plus manual comparison with generated and training images.

BOUNDARY

These associations suggest failure modes and dataset issues; they do not establish causal mechanisms.

[1] §4.3.3, Figs. 8–9 and Table 7 · pass rates span approximately 93.5%–99.5%

The paper demonstrates feasibility—not a complete assurance process.

GENERATIVE TRAINING

Better tests

Improve generator training and adaptation to reveal more observed failures and reduce false positives.

COVERAGE EVIDENCE

Latent coverage

Adapt systematic latent-space coverage when test generation does not reveal a failure.

GENERALIZATION

More domains

Expand driving-like safety datasets and explore broader robustness and functional requirements.

ASSURANCE BOUNDARY

Automated labels, selected datasets, narrow baselines, and small KID subsets limit how far the evidence generalizes.

[1] §4.4 and §5 · author-stated scope kept separate from this seminar's critique

The paper jointly validates 330 sampled failures—but not the full failure population.

PRECONDITION INVALID · MODEL PASSES **Irrelevant pass** The output says nothing about the stated requirement.	PRECONDITION INVALID · MODEL FAILS **Off-target false positive** A model error is not yet a requirement counterexample.
PRECONDITION VALID · MODEL PASSES **Valid requirement pass** Relevant evidence, but never proof of the universal property.	PRECONDITION VALID · MODEL FAILS **Credible counterexample** This joint event is the paper's strongest testing evidence.

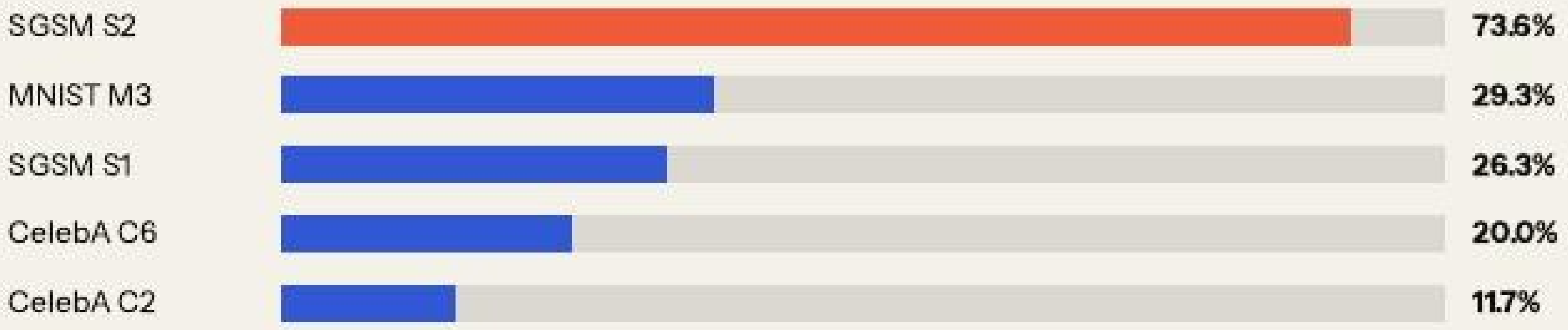
SAMPLED JOINT AUDIT

30 failures × 11 low-pass requirements = 330 independently assessed samples.

Fair critique: the paper did perform joint validation. Its generalization remains limited by author assessment, fixed samples, no reported confidence intervals, and no external or blinded annotators.

The proxy and 42-image LLM check identify audit targets; neither estimates population validity.

Independence-proxy ranking



Diagnostic only · unconditional P(match) × P(fail) · not an observed joint rate

SMALL EXTERNAL SEMANTIC CHECK

.738

31 of 42 sampled images were judged precondition-valid by one multimodal model.

Use: decide where independent joint review is most valuable.

The product assumes independence. The paper's conditional human audit among failures is methodologically stronger; this LLM check is only a second opinion.

Yes—but our shared VAE asks a different, still unresolved question.

PAPER · ALL-DATA LORA

One FLUX LoRA sees every MNIST training image.

It is tested in RQ1, misses rare preconditions, and is dropped from later RQs.

M3 match · 7.4%

SEMINAR · SHARED VAE

One new conditional VAE sees copied RBT4DNN outputs.

Its digit-pass pattern is similar on MNIST, but it is neither FLUX nor LoRA and lacks RQ1–RQ3 validation.

mean pass · .945

NEXT CONTROLLED STUDY

One shared conditional FLUX adapter on original data.

Compare with per-precondition LoRAs using the same prompts, GTCs, KID, JS, and independent human audit.

open question

WHY SCALABILITY MATTERS

21 primary adapters × reported 52.1–82.5 minutes each

≈24.7 serial A100-hours

[1] Fig. 3 and artifact · seminar mnist-shared-generator · total excludes unreported workflow costs

Requirements make tests relevant. Validity makes failures credible.

WHAT THE PAPER ESTABLISHES

Structured requirements can guide distributionally plausible, glossary-diverse, scenario-specific test generation across four domains.

WHAT REMAINS OPEN

Broader independent joint audits, shared generators, alternative adaptation methods, and end-to-end replication at larger scale.

References.

[1] **N. J. Mozumder, F. Toledo, S. Dola, and M. B. Dwyer.** “RBT4DNN: Requirements-based Testing of Neural Networks.” arXiv:2504.02737, v3, 2025.

[2] **E. J. Hu et al.** “LoRA: Low-Rank Adaptation of Large Language Models.” ICLR, 2022.

[3] **RBT4DNN artifact.** github.com/less-lab-uva/RBT4DNN · seminar provenance pinned to commit c3deff871ed41043d83a05289faab84c41706586.

[4] **Seminar experiments.** Reproducible notebooks, results, summaries, and provenance in this repository.

Original paper ↗	Original artifact ↗